

A Security System for Door Opened by Using Brain Signal

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Abstract - A Brain computer interface (BCI) has introduced a new zone for inventive applications for researchers and designers offering various channels of communication to those humans who are suffering of motor disabilities. Most of existing secure door opened systems are based on sensor and alarms. A possible solution is the use brain computer interface system (BCI). In addition, it is important to study the BCI framework. In this research, a security system for door opened by using brain signal framework developed based on time-frequency techniques is proposed. The time-frequency method for feature extraction namely empirical mode decomposition (EMD) was combined separately with different classification methods such as K nearest neighbours (KNN), liner discriminate analysis (LDA) and Support vector machine (SVM). To validate this framework, experimentally recorded data from nine subjects were collected. The feature extraction-based method EMD with classification SVM obtained average accuracy of 84%. The proposed framework based on SVM is more precise resulting in the improvement of accuracy, compared to other classification methods.

Index Terms - Brain-computer interface (BCI), EEG, Motor imagery, Empirical mode decomposition (EMD), liner discriminate analysis (LDA), Support vector machine (SVM) and K nearest neighbors (KNN).

I. INTRODUCTION

The brain- computer interface (BCI) has declare a new dimension and created a new era for **inventive** applications. BCI have provided a subtitle form of communication to control wheelchair, robot, security system, and act. The signal detected through electroencephalography (EEG) when they are modulated and use as input for door opening. This is because EEG has high temporal resolution and probability [1]. There are several techniques to measure brain signals. We can divide them into Non-Invasive, which is the EEG signal is taken placing electrodes on the scalp, so on the most external part [2]. Semi-invasive which is the ECoG signal is taken from electrodes placed in the Dura or in the arachnoid and Invasive. Invasive, which is the Intraparenchymal signal is taken directly implanting electrodes in the cortex. An significant goal of pre-processing is to enhance the quality of signal by enhancing the signal-to-noise ratio (SNR) [3]. Selecting the appropriate features affects the recognition rate and the efficiency of the BCI

system [4]. Different features extraction techniques had been attempted to spot the mental tasks motor imagery, such as fast Fourier transform (FFT) [5], power spectral density (PSD) [6], autoregressive filter [7], adaptive autoregressive filter (AAR) [8], common spatial patterns (CSP) [9] and band power (BP) [10]. Despite the large number of features that were utilized in the previous BCI applications, they do not satisfy the application of EEG-based BCI system [11]. This is because these techniques are based on time or frequency domain. Therefore, it is essential to develop efficient features extraction methods that are robust and consider non-stationary phenomena of EEG signals. Empirical mode decomposition (EMD) carries on a conceptual advance in time-frequency analysis for non-stationary and non-linear signals [12] [13]. The purpose of this study is design a Security system based Brain signal for door opening [5]. The framework is based on time-frequency feature extraction technique. Which is empirical mode decomposition (EMD) to discriminate mental tasks motor imagery. This is due to their capability to provide the required signal, which can be used for real-time experiments on **door** opened. KNN, LDA and SVM are applied for feature classification.

II. MATERIAL AND METHODS

A. Participant and Experimental Setup

The experiment session performed mental task thinking was shown for participants. The participants of this experiment were undergraduate students from University of Khartoum. The number of the participants were 9 subjects (4 males) and (5 females), with a mean age of 24.5 years, and standard deviation of 4.1. The experimental setup consisted of Muse headset, it is a lightweight, rechargeable headband. It has 4 EEG electrodes: two on the forehead, slightly above the eyes, and two touching the ears. Additionally, it also has a gyroscope and accelerometer, so you can calculate the head orientation 2016 Headband EEG system. The EEG sensor positions were TP9, AF7, AF8, TP10. The lobe of the right ear utilized for the signal reference and ground electrode. The sampling frequency was 500Hz [5] as shown in Fig. 1

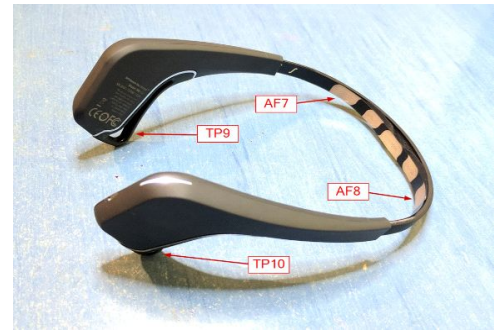


Fig. 1 EEG Muse device

B. Experimental Session

As shows in Fig. 2 the experimental session performs mental task thinking. Through the session, the screen of the participant might be blank or could show hello or bad (represents a specific password). Each stimulus is shown continuously for 3 seconds, however when time is over, the screen will keep on blank for 1 second before the next stimulus appears. Therefore, between each run, there is a break of 1 minute while the screen is blank and the subject might blink or stretch.

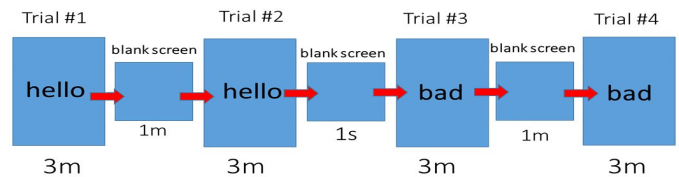


Fig. 2 Experimental session protocol

Each subject was instructed to imagine "hello" or "bad" as a password. Each participant performed for a "hello" 15 times and "bad" 15 times runs, with each run containing of 30 trials; within these 30 trials per run. In total for the 15 runs, the overall trials per subject were 270, which corresponding to the different thinking tasks performed in a random order. In further, the overall number of trials per task per subject were 30.

C. Feature Extraction

The feature extraction is investigated the most appropriate information for classifying the EEG patterns. The main objective of this stage is to advance the performance of the classification of the BCI system and it is typically performed along with the classification stage. In this work we used Empirical Mode Decomposition (EMD) carries on a

conceptual advance in time-frequency analysis for non-stationary and nonlinear signals. Any EEG data can be decomposed into numerous independent intrinsic mode functions (IMFs). A signal $x(t)$ into intrinsic mode functions $c_i(t)$, for $i = 1, 2, \dots, n$ and residue $r(t)$:

$$x(t) = \sum_{i=1}^n c_i(t) + r(t)$$

(1)
where n is the mean number of IMF functions. The residue $r(t)$ reflects the average trend of the signal $x(t)$ or a constant value.[6]

D. Classification

The features extracted step is the input for the classifier algorithm. However, the classifier could compute the potentials for the class's inputs and it will choose the one with the highest opportunity at this step. In this paper we used a ready tool in Matlab that provides and facilitates the classification process, and this tool contains many classification algorithms built inside it. We used linear discriminant analysis (LDA), support vector machine (SVM) and K nearing neighboring (KNN) classifiers.

III. RESULTS AND ANALYSIS

From the experimental, the data was classified offline, the majority of the subjects obtained an accuracy of 53.3 – 100 %. However, accuracy values lower than 80% were observed in five subjects. The mean accuracy average over the subjects was approximately 84% using SVM. Table 1 shows the classification accuracy obtained from three classifiers.

Table 1 Results obtained the classification of participants

Subject	LDA	SVM	KNN
S1H	96.7%	100%	96.7%
S2A	93.3%	90.0%	90.0%
S3B	93.3%	93.3%	96.7%
S4AN	90.0%	90.0%	86.7%
S5G	70.0%	80.0%	76.7%

S6M	66.7%	60.0%	53.3%
S7MO	70.0%	90.0%	80.0%
S8Y	80.0%	73.3%	80.0%
S9R	63.3%	76.7%	66.7%
Mean	83.37%	84%	80.76%

The effects of different classifiers and subjects performance on the accuracy are presented in Fig.3. The accuracies obtained were 100% with SVM 96.7% with LDA and 96.7% with KNN for S1, for S2 were 90.0% with SVM, 93.3% with LDA and 90.0% with KNN, for S3 were 93.3% with SVM, 93.3% with LDA and 96.7% with KNN and for S4 were 90.0% with SVM, 90.0% with LDA and 86.7% with KNN.

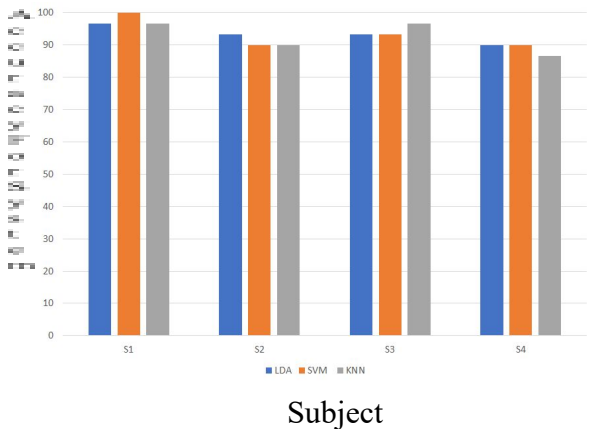


Fig.3 The comparisons among the LDA, SVM and KNN.

The confusion matrix of the two classes using SVM classifier is presented in Fig.4. The true class of bad is counted of 9 and were misclassified by 4. Hello class is counted of 15 true class and were not misclassified.

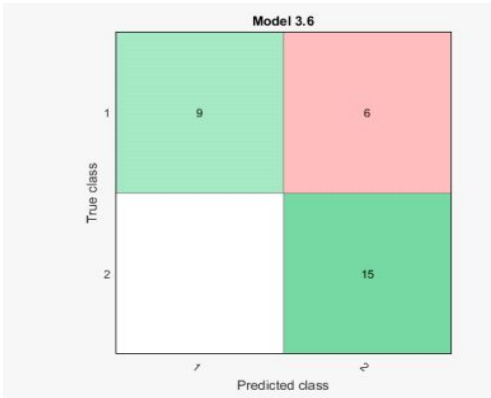


Fig. 4 Confusion matrix for hello and bad using KNN.

The receiver operating characteristic (ROC) curves of the two class motor imagery using SVM classifier is presented in Fig. 5.

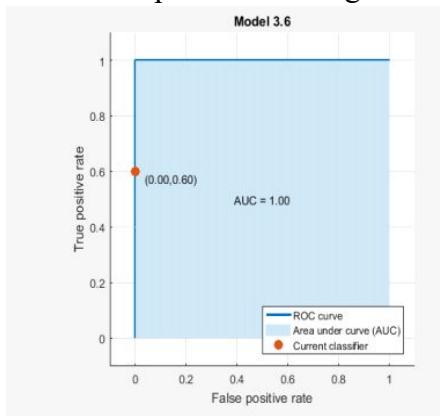


Fig. 5 The Roc curve for hello and bad using SVM.

IV. CONCLUSION

This study pointed to develop a proper classifier for discrimination of two classes motor imagery tasks for a security system for door opened. EMD based decomposition technique was used. The system was successfully built and trained offline, the password was recognized successfully. In addition, a comparison among the three classifiers showed that SVM outperformed LDA and KNN with an average accuracy of 84%, 83.37% and 80% respectively

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